SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Based Questions

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4-AT3 Viva Based Questions**	**Assessment:**	**ASSESSMENT TOOL 3 – Viva Based Questions**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
Student Name:	Sania Herbert	**Reg. No.:**	192324062
Section / Batch:	2023/AI&DS	**Date:**	23.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

C. Point Estimation

Que:1 What is a point estimate?
Ans: A point estimate is a single numerical value calculated from sample data that is used to estimate an unknown population parameter.
For example, if the sample mean transaction value of 200 customers is ₹18,500, then ₹18,500 is the point estimate of the population mean.
Formula: ^μ=x¯
where:

- **X-= sample mean**
- **μ= point estimate of population mean**

que:2 What is an estimator?
An estimator is a statistical rule or formula that uses sample data to estimate an unknown population parameter.
For example, the sample mean is an estimator of the population mean**.**

$$\bar{X} = \frac{\sum X_i}{n}$$

Here, X¯ is the estimator, because it is a formula that can be applied to different samples.

Que:3. What is the difference between an estimator and an estimate?

Estimator	**Estimate**
It is a formula or statistical rule.	It is the numerical value obtained from the estimator.
It is based on random sample data.	It is based on a particular sample.
Example: (\bar{X})	Example: ₹18,500
It can vary from sample to sample.	It is a specific calculated value.

Example:

Suppose the sample mean formula is:

$$\bar{X} = \frac{\sum X_i}{n}$$

This formula is the estimator.
If the calculation gives:
$\bar{x}$=18,500 then ₹18,500 is the estimate.

Que: 4. What is the point estimator for the population mean?
The sample mean is the point estimator for the population mean.

$$\boxed{\hat{\mu} = \bar{X} = \frac{\sum X_i}{n}}$$

where:

- **Xi= individual sample observations**
- **n= sample size**
- **X-= sample mean**

For example, if 200 customers have an average transaction value of ₹18,500, then: $\hat{\mu} = ₹18,500$

Thus, the sample mean provides the point estimate of the population mean.

Que:5. How do you estimate a population proportion from a sample?
A population proportion is estimated using the sample proportion.
The formula is:

$$\hat{p} = \frac{x}{n}$$

where:

- **x= number of individuals having the required characteristic**
- **n= total sample size**

- **P^= sample proportion**

Example:
Suppose 80 out of 200 customers use mobile banking.
p^=80/200=0.40
Therefore:

$$\hat{p} = 0.40 = 40\%$$

Hence, 40% is the point estimate of the population proportion.

Que:6. What is the point estimate of population variance?
The sample variance is commonly used as the point estimator of population variance.

The unbiased sample variance is: $$s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

where:

- **s^2= sample variance**
- **X-= sample mean**
- **n= sample size**

For example, if the calculated sample variance is: s2=17,640,000 then 17,640,000 is the point estimate of the population variance.
Important: The denominator is n-1 , not n , when using the usual unbiased estimator of population variance.

Que:7. Why is the sample mean commonly used to estimate the population mean?
The sample mean is commonly used because it has several desirable statistical properties:

1. **Unbiasedness:**
   The expected value of the sample mean equals the population mean. E(X¯)=μ
2. **Consistency:**
   As the sample size increases, the sample mean tends to get closer to the population mean.

3. **Efficiency:**
   Under common assumptions, the sample mean has relatively low variance among unbiased estimators.
4. **Easy to calculate:**
   It is simple to compute and understand.
5. **Uses all observations:**
   Every observation in the sample contributes to the sample mean.

Therefore, the sample mean is a reliable and widely used estimator of the population mean.

Que:8. What properties should a good estimator have?

A good estimator should ideally have the following properties:

1. Unbiasedness

The expected value of the estimator should equal the true population parameter. E(θ^)=θ

2. Consistency

The estimator should approach the true population parameter as the sample size increases.

3. Efficiency

Among unbiased estimators, a good estimator should have a smaller variance.

4. Sufficiency

The estimator should use all the relevant information contained in the sample about the parameter.

Therefore:

$$\text{Good estimator} \rightarrow \text{Unbiased} + \text{Consistent} + \text{Efficient} + \text{Sufficient}$$

Que:9. What is an unbiased estimator?

An unbiased estimator is an estimator whose expected value is equal to the true population parameter.

If estimates the population parameter , then: $E(\hat{\theta}) = \theta$

For example, the sample mean is an unbiased estimator of the population mean: E(X¯)=μ

This means that if we repeatedly take samples and calculate their means, the average of those sample means will equal the true population mean.
Example:
If the true population mean is ₹20,000, an unbiased estimator does not systematically overestimate or underestimate ₹20,000.

Que:10.What is meant by the bias of an estimator?
Bias measures the systematic difference between the expected value of an estimator and the true population parameter.

The formula is: $Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$
where:

- E(θ^)= expected value of the estimator
- θ= true population parameter

Example:
Suppose: E(θ^)=105
and the true parameter is: θ=100
Then: Bias(θ^)=105−100=5
Therefore, the estimator has a positive bias of 5.
If: Bias(θ^)=0
the estimator is unbiased.

Que:11. What is the difference between bias and variance of an estimator?
Bias and variance measure different aspects of an estimator.

Bias	Variance
Measures systematic error.	Measures variability of the estimator.
Shows how far the expected estimate is from the true parameter.	Shows how much estimates vary between different samples.
Formula: E(θ^)−θ	Formula: E[(θ^−E(θ^))2]
Bias can be positive, negative, or zero.	Variance is always non-negative.

Simple example:
Suppose we repeatedly take samples from the same population.

- If the estimates are consistently away from the true value, there is bias.
- If the estimates are widely spread out, there is high variance.

A good estimator generally aims for low bias and low variance.

Que:12. Why can two different samples produce different point estimates for the same population parameter?

Two different samples can produce different point estimates because of sampling variability.
A sample represents only a portion of the population. Different random samples may contain different observations, resulting in different sample statistics.
For example:
Sample 1: $\bar{x}_1$=₹18,200
Sample 2: $\bar{x}_2$=₹18,700
Both samples are estimating the same population mean, but they produce different estimates.
This difference occurs because of random sampling variation.
As the sample size increases, the estimates generally become more stable and tend to get closer to the true population parameter.